

VNU-HUS MAT1206E/3508: Introduction to AI

Introduction In-class Discussion

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What is Artificial Intelligence (AI)?

What this section covers



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There is no single agreed definition of AI, so we approach it from several angles:

- *Classic definitions* (McCarthy, Encyclopedia Britannica, Elaine Rich, Turing) — and why none is fully satisfactory.
- *A framework* that organises them: Russell & Norvig's *four approaches* and the *rational agent*.
- *The Turing test* — and whether today's large language models pass it (and what that really shows).
- *Searle's Chinese Room*: behaving intelligently vs. genuinely understanding.
- *Weak/strong* and *narrow/general (AGI)/super (ASI)* AI.
- *AI today*: foundation models, generative AI, and agentic AI.

Goal

By the end you should be able to *place any system* on these maps and argue *how you would define AI*.

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What is Artificial Intelligence (AI)?

At a very first glance,

- Humans have *real* intelligence
- Machines have *artificial* intelligence

Basic Idea

Artificial intelligence is based on the assumption that the process of human thought can be *mechanized*

- Human thought can be described as a mechanical sequence of steps that machines can execute.
- This implies that machines can be designed to simulate human cognitive processes, such as *adaptation, learning, and decision making*.

Note

Indeed, *there is no officially agreed definition of AI*. It is thus not easy to categorize what is AI and what is not

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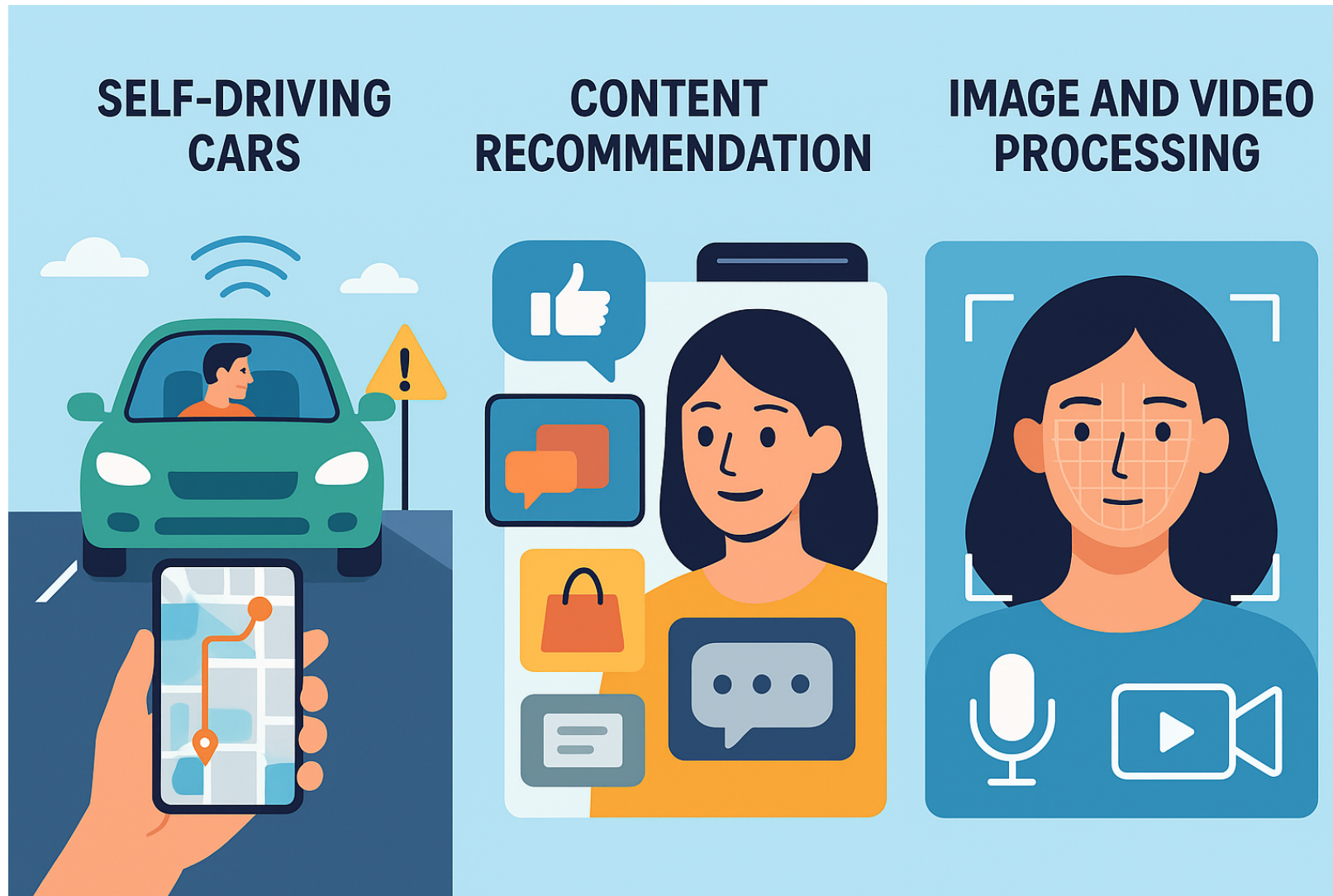


Figure: Some real-world applications which combine several “AI-techniques”. Image generated by ChatGPT

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What do you think?



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John McCarthy (1955) [one of the pioneers of AI]

The goal of AI is to develop machines that behave as though they were intelligent

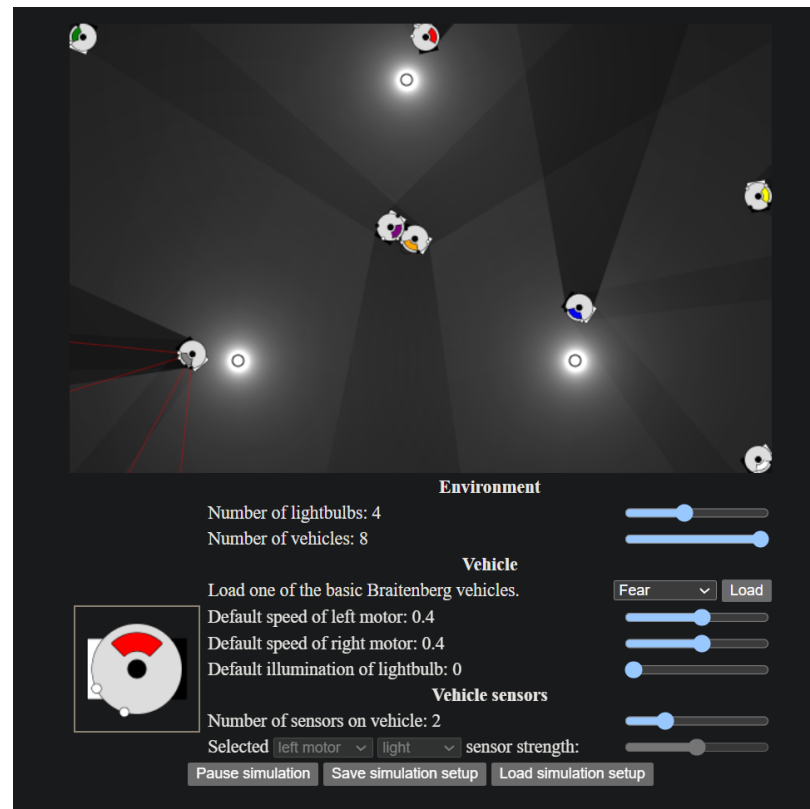


Figure: Example of Braitenberg vehicles simulating simple behaviors.
(See <http://www.harmendeweerd.nl/braitenberg-vehicles/>)

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Encyclopedia Britannica (1991)

AI is the ability of digital computers or computer controlled robots to solve problems that are normally associated with the higher intellectual processing capabilities of humans ...

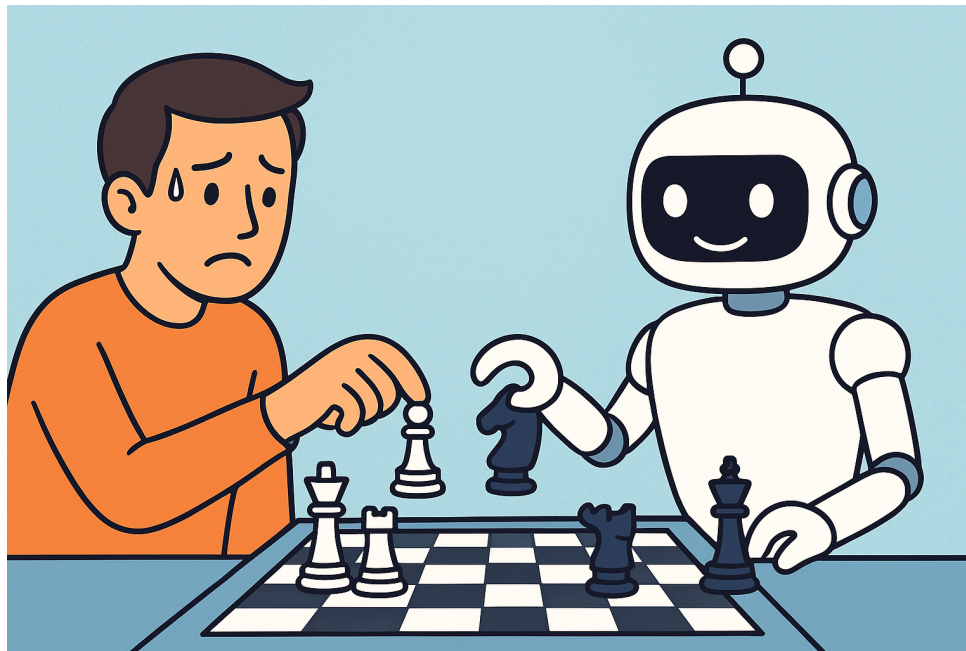


Figure: Computers can now beat humans in complex games (e.g., chess, Go), yet robots still struggle with dexterous tasks such as grasping and precise object placement — an instance of *Moravec's paradox* (1988). Image generated by ChatGPT

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Elaine Rich [Rich 1983]

Artificial Intelligence is the study of how to make computers do things at which, at the moment, people are better



Figure: AI systems can now outperform humans in tasks such as playing chess, yet they still have limitations in true understanding and general reasoning. (Image generated by Grok)

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Alan Turing [Turing 1950]

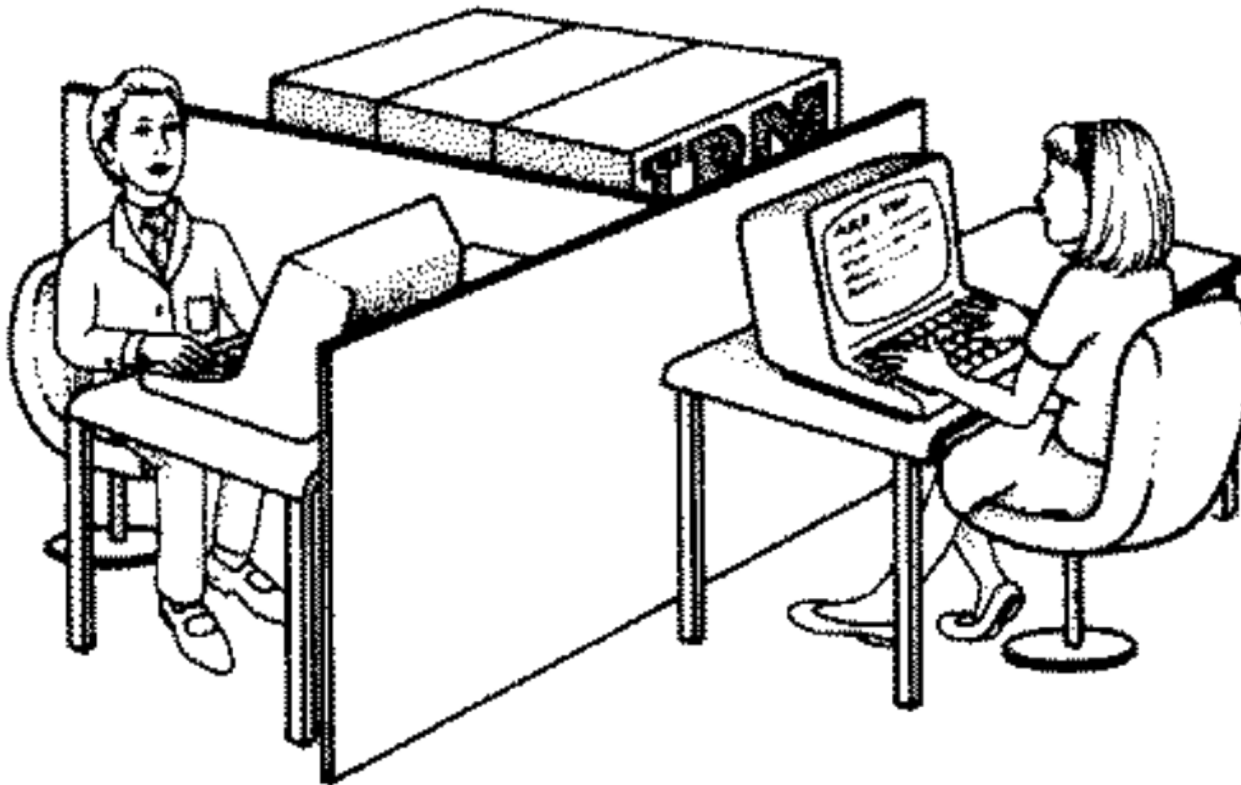


Figure: Turing test: after a five-minute conversation, the machine passes if it can mislead Alice in *at least* 30% of the cases.

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A framework: the four approaches



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Russell & Norvig [Russell and Norvig 2021] organise definitions of AI along two axes — *thought vs. behaviour* and *human-like vs. rational* (i.e. *goal-optimal*):

	Humanly	Rationally
Thinking	cognitive modelling	laws of thought (logic)
Acting	<i>the Turing test</i>	<i>the rational agent</i>

Here *rational* means “*doing the right thing*”: pick the action expected to best achieve the goal — *not* “human-like”, and not merely “logical”.

- In Russell–Norvig’s table, these definitions fit best under *acting humanly*: McCarthy and Turing emphasize human-like behaviour, while Britannica and Rich use a broader human-performance comparison, judging whether a system can solve tasks normally associated with human intelligence rather than whether it imitates a whole person.
- Some AI courses adopt *acting rationally*: design an *agent* (= something that acts) that perceives its environment and acts to achieve the *best expected outcome*.
- Many ML/deep-learning–focused intros instead define AI as *learning from data*.

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More on Turing Test



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The idea that *“intelligence is the same as intelligent behavior”* has been challenged by some. The best known counter-argument is John Searle’s *Chinese Room Argument* (1980)

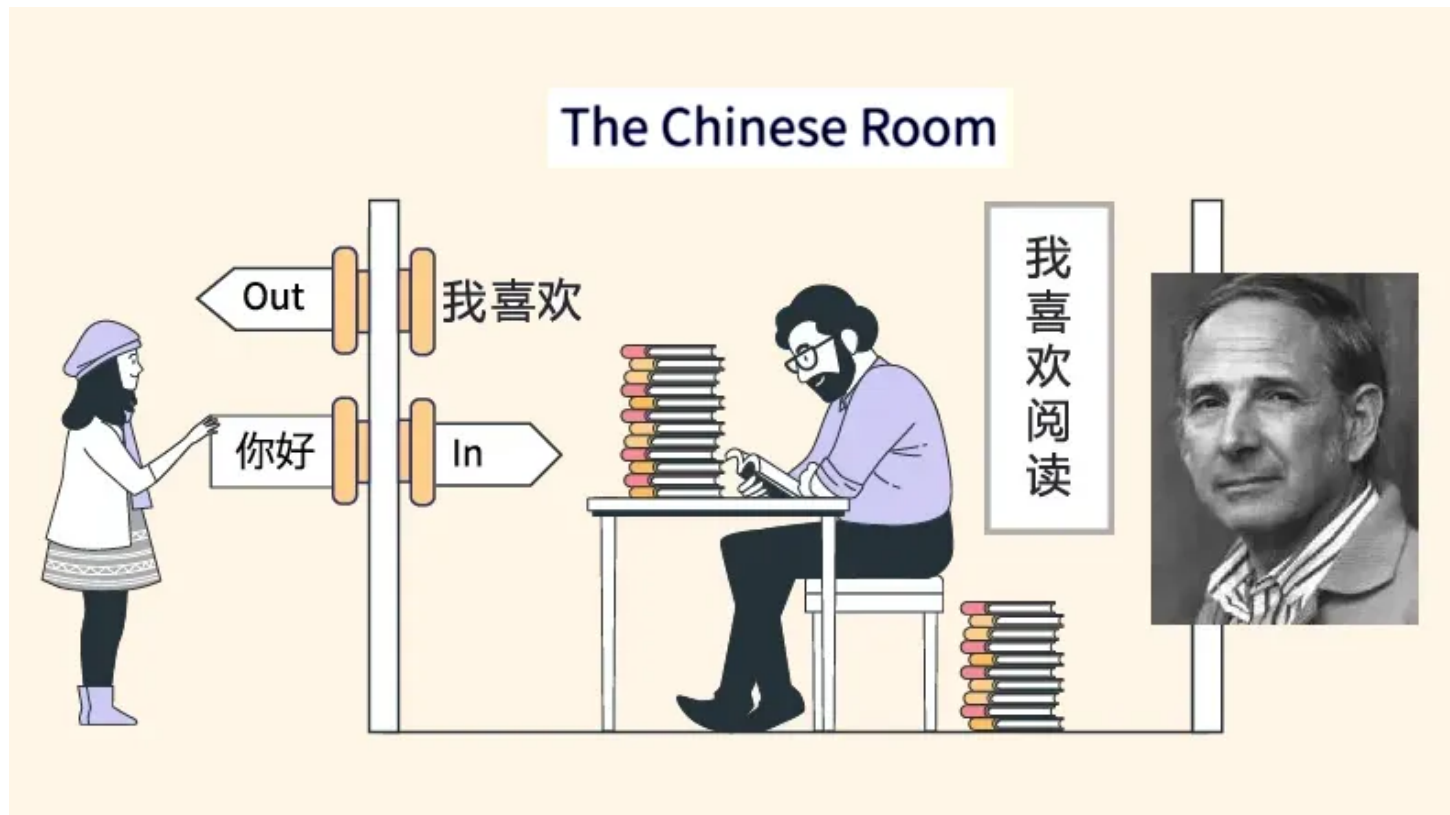


Figure: The Chinese Room Argument by John Searle (1980). Image taken from the Internet



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Two different distinctions are often confused here:

- *Does the machine really think? (Searle, 1980)* Under *strong AI* an appropriately programmed computer *literally has a mind* and genuinely understands; under *weak AI* it only *simulates* thinking and is a useful tool. Searle's Chinese Room argues against strong AI.
- *How broad are its abilities?* *Narrow AI* is built for one specific task (e.g., facial recognition, web search, driving); *general AI (AGI)* could perform any intellectual task a human can. *Most current AI is narrow AI.*

Additionally, the strong development of current AI technologies suggests the idea of *artificial superintelligence (ASI)*, a level of intelligence that surpasses human intelligence across all domains. (For example, see <https://www.ibm.com/think/to-pics/artificial-superintelligence>)

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What “AI” means today



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Today's AI is dominated by systems *learned from data*, often described in three layers:

Foundation models / LLMs huge models pre-trained on massive data (GPT, Claude, Gemini, Llama).

Generative AI foundation models used to *produce* text, images, audio, or code from a *prompt* (ChatGPT, Midjourney).

Agentic AI models that *plan, call tools, and act autonomously* over many steps. They mainly show *autonomy*; some are also *adaptive* when they improve from feedback or experience.

Nesting: *deep learning* $\subset$ *machine learning* $\subset$ *AI* — *ML is a subfield, not a synonym for AI.*

Caveat

Fluency hides real limits: *hallucination*, brittle reasoning, high cost, bias, and *no guarantee of understanding*.

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So... is it AI? A practical test



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With no agreed definition, some suggest testing two properties:

- *Autonomy* — performs tasks in complex environments without constant guidance.
- *Adaptivity* — improves performance by learning from experience.

Exercise 1 (Border cases)

Score each on autonomy and adaptivity — is it “AI”?

(1) a thermostat, (2) a spreadsheet computing a sum, (3) a GPS route-planner, (4) a spam filter, (5) ChatGPT

Two cautions

AI $\neq$ *robotics*: the intelligence is in the decision-making, not the hardware (robots may *use* AI). And *intelligence/understanding* are “*suitcase words*” (Minsky) — they bundle many meanings, fuelling hype.

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Exercise 2

There are a few other ways people define AI, for example,

- AI is any “cool things that computers can do”
- AI is any machine which simulates human behavior
- AI is an autonomous¹ and adaptive² system

Among all definitions we have mentioned, which one do you like *best*? How would *you* define AI?

Exercise 3

List some real-world applications of AI which we have not yet mentioned.

¹The ability to perform tasks in complex environments without constant guidance by a user.

²The ability to improve performance by learning from experience.

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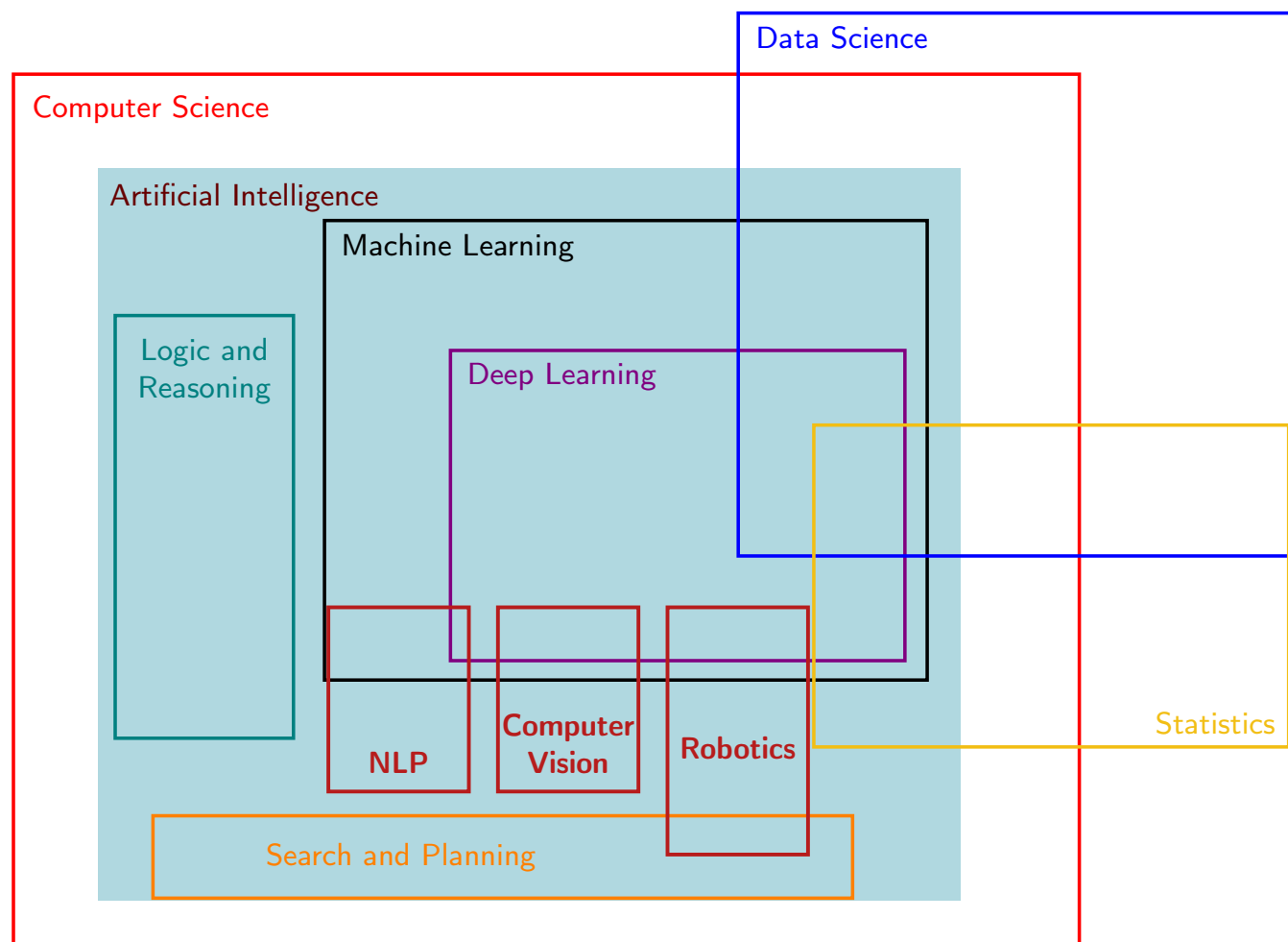


Figure: Related fields of AI. A box *inside* another means “is a subfield of” ($CS \supset AI \supset ML \supset DL$); *overlapping* boxes share methods, neither containing the other. NLP, computer vision and robotics *straddle* ML (their modern methods are deep-learning-based); robotics also overlaps search & planning. *Non-touching boxes are weakly, not un-, related.*

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Early beginnings and the symbolic era



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Note: There are different ways to present the history of AI. Here is a reasonable summary.

Two lenses: AI swings between *symbolic* and *connectionist* approaches, and moves in *hype cycles*: bold predictions → disappointment → “*winters*” → revival.

■ Early beginnings (1930s–1950s)

- Theoretical foundations: formal logic, computation theory (Church–Turing), early automata.
- McCulloch & Pitts (1943): first mathematical model of a neural network. Turing (1950): the Turing test.

■ Birth and the symbolic (“logic”) era (1956–1974)

- Named at the *1956 Dartmouth Conference* (McCarthy coins “artificial intelligence”); Newell & Simon’s *Logic Theorist* (the first automated theorem-proving program); McCarthy’s programming language *LISP* (1958).
- Bold predictions: Newell & Simon (1958) — a chess world-champion computer “*within ten years*” (actually *1997*, Deep Blue — ~40 years late); Simon (1965) — machines doing “any work *that* a man can do” within twenty years.

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First AI winter and expert-systems boom



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■ First AI winter (1974–1980)

- Over-promising met reality — the *combinatorial explosion* (the number of possibilities to search grows so fast that early methods could not scale to real problems). The *Lighthill Report* (1973), a UK government review judging that AI had over-promised, plus DARPA funding cuts, triggered the winter. Earlier, Minsky & Papert's book *Perceptrons* (1969) had stalled neural-net research by proving that a single-layer network can't even learn *XOR* (a simple either/or logic function).

■ Expert-systems boom (1980–1987)

- Industry adopts rule-based *expert systems* — programs that encode a human expert's *if-then rules* (MYCIN diagnosed infections, DENDRAL identified molecules, XCON/R1 configured computer orders — saving DEC ~\$40M/year). Japan launches the *Fifth Generation* project (1981), a national push to build AI-oriented computers.

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Second AI winter and statistical revival



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■ Second AI winter (1987–1993)

- The market for *LISP machines* (specialised computers built to run AI programs) collapses; expert systems prove *brittle* and costly — the *knowledge-acquisition bottleneck*: every rule had to be hand-coded from human experts, which was slow and did not scale. To shed the “AI” stigma, researchers rebrand as *machine learning*, *pattern recognition*, *informatics*.

■ Statistical revival and consolidation (1990s–2000s)

- The field turns to *probability and data* (Bayesian networks, support-vector machines — methods that *learn patterns from data* instead of following hand-coded rules) and ships AI in real products: IBM’s chess computer *Deep Blue* beats *Garry Kasparov* (1997), then the reigning *world chess champion* — the first time a machine beat a world champion at chess; IBM’s question-answering system *Watson* wins the US TV quiz show *Jeopardy!* (2011), beating its best human champions.



■ Deep-learning breakthrough (2012–2016)

- the deep neural network *AlexNet* (2012) wins the *ImageNet* contest — a yearly competition to sort over a million photos into 1000 categories; AlexNet roughly *halved the error rate*, proving deep learning works and *kicking off* today's deep-learning era. DeepMind's Go program *AlphaGo* beats *Lee Sedol* (2016), a top human champion at *Go* — a board game long thought far harder for computers than chess. Enabled by *big data* and *GPUs*.

■ Generative AI and foundation models (2017–present)

- The *Transformer* architecture (2017) — which reads a whole sequence at once using “*attention*” — underlies today's *large language models*: GPT-3 (2020) → *ChatGPT* (Nov 2022) → GPT-4 (2023) → reasoning/agent models (2024–25). DeepMind's *AlphaFold 2* (2020/21) solves *protein-structure prediction* — computing a protein's 3D shape from its amino-acid sequence, a ~50-year grand challenge in biology (2024 Nobel in Chemistry). AI becomes pervasive — raising social and ethical questions.

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Why AI is “hot” today?



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■ It really works now — and it’s everywhere

- Today’s AI skips the big philosophical questions and just *solves real, useful problems*.
- It is already in tools you use daily (phone assistants, recommendations, ChatGPT), and runs for real in healthcare, finance, and education — not only in labs.

■ We finally have the tools to build it

- The ingredients are all here — *lots of data*, *fast chips* (GPUs/TPUs), *cloud computers*, and *free open-source software* — so building and shipping AI is now fast and cheap; almost anyone can try.

■ But stay a little skeptical

- AI has been over-hyped before: bold promises (“human-level AI in ten years”) kept falling short. Enjoy the progress, but weigh today’s excitement against that track record.

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Exercise 4

Do your own research on the history of AI. Some interesting topics would probably be:

- (a) The *Dartmouth Conference* and the birth of AI
- (b) The rise and fall of *expert systems*
- (c) The history and impact of *neural networks and deep learning* on AI (recall: *machine learning* is a subfield of AI, and *deep learning* is a subfield of machine learning)
- (d) The history and development of *two major theories of AI*: *symbolic AI* (AI should be based on traditional computer logic or preprogramming $\Rightarrow$ Modeling human's *mind*) and *connectionist AI* (AI needed to use systems similar to the brain like neural networks $\Rightarrow$ Modeling human's *brain*). (Some scientists want to go further by blending the two approaches into something called *neuro-symbolic AI*.)
- (e) The *"AI effect"*: AI is a moving target — once a problem is solved (chess, OCR, translation), people stop calling it "AI". How might this help explain the field's repeated "winters" and rebranding?

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How to think about AI's impact



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AI is just a tool: its effects depend on how we *build, use, and govern* it. To judge any application, ask which *values* it affects:

- *Fairness* (does it discriminate?), *Transparency* (can we explain or contest it?), *Accountability* (who is responsible?), *Privacy*, *Safety & security*, and effects on *work and society*.

Each implication below stresses one or more of these. We look at the *risks*, then the *benefits*, then what is being *done*.

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Algorithmic Bias



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The screenshot shows a legal insights article from Holland & Knight, dated May 27, 2025. The title is "Federal Court Allows Collective Action Lawsuit Over Alleged AI Hiring Bias". The subtitle is "Workday Inc. Now Faces Nationwide Collective Action Over Alleged Age Discrimination in Artificial Intelligence (AI) Hiring Tools". The article is categorized as a "Holland & Knight Alert" and is authored by Jennifer Lada, Bryan Neal, D'Andre Chapman, and Laura Askinazi. The main text describes how Derek Mobley, a 40-year-old man, sued Workday Inc. claiming that the company's AI-driven applicant screening tools systematically disadvantaged him and other older job seekers. Mobley submitted more than 100 applications to companies using Workday's platform, but he was rejected every time. He alleged that the AI tools, designed to score, sort, rank, or screen applicants, unfairly penalized older candidates. The court initially dismissed Mobley's complaint but granted permission to file an amended version. Subsequently, Workday attempted to have this amended complaint dismissed as well but was unsuccessful. The court, by denying Workday's motion to dismiss, recognized Mobley's claim as plausible under the Age Discrimination in Employment Act (ADEA), based on a disparate impact theory. This development marks a pivotal moment in the ongoing legal proceedings, as it allows Mobley to continue pursuing the case against Workday. The article also includes a section on "Key Developments" dated May 16, 2025, where Judge Rita Lin of the U.S. District Court for the Northern District of California granted preliminary certification under ADEA, allowing the lawsuit to move forward as a nationwide collective action – similar to a class action but with more flexibility. A sidebar on the right lists "RELATED BLOG" (Cybersecurity and Privacy Blog) and "RELATED PRACTICES" (Labor and Employment, Labor and Employment Class Actions, Civil Rights, Discrimination and Retaliation, Artificial Intelligence, Artificial Intelligence Policy & Regulation, Technology, Emerging Technologies Law and Policy).

Figure: *Mobley v. Workday* (2025): a man rejected by AI hiring software (from the company *Workday*) is suing, saying it *unfairly rejected him because of his age*. A U.S. court let others rejected the same way *join the case* — which asks: when an AI turns you down for a job, *who is to blame?*

(<https://www.hklaw.com/en/insights/publications/2025/05/federal-court-allows-collective-action-lawsuit-over-alleged>)

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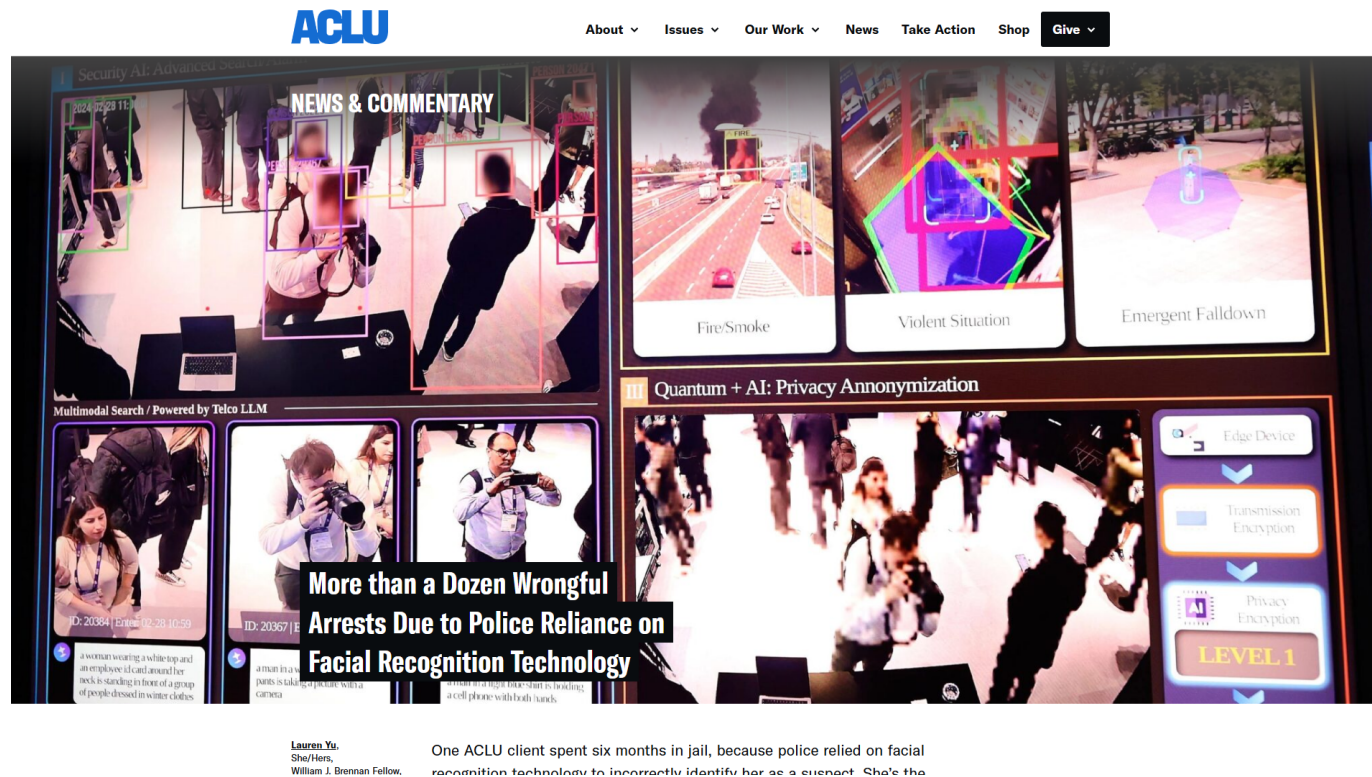


Figure: ACLU (2026): a *14th* documented wrongful arrest from police reliance on facial recognition — the technology flags innocent people who merely *look similar*, and the harm falls overwhelmingly on Black Americans. (<https://www.aclu.org/news/privacy-technology/more-than-a-dozen-wrongful-arrests-due-to-police-reliance-on-facial-recognition-technology>)

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Figure: Stanford “Algorithmic Monocultures in Hiring” (2026): when *many employers all use the same* AI screening tool, that one tool’s small bias gets repeated everywhere — so the *same people keep being rejected* (Black and Asian applicants more often).

(<https://digitaleconomy.stanford.edu/publication/algorithmic-monocultures-in-hiring/>)

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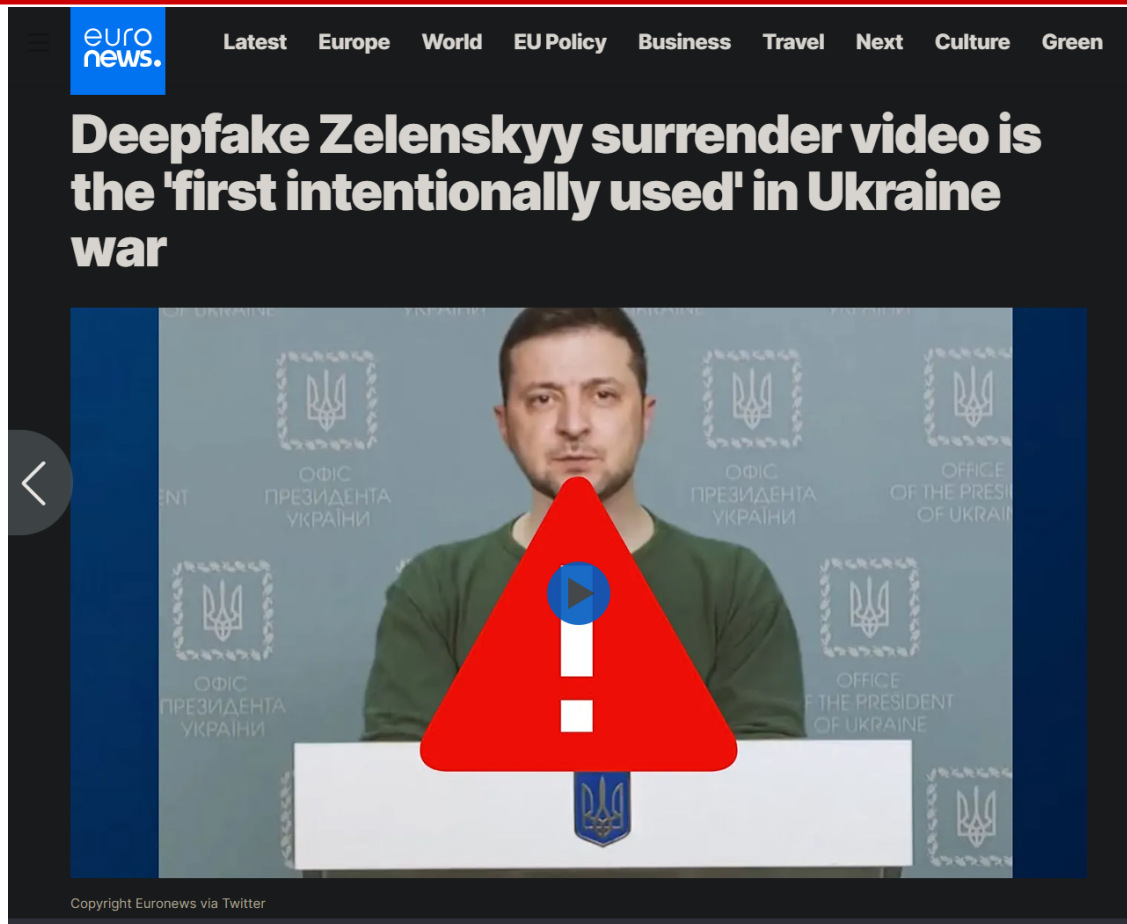
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- Algorithmic bias isn't a hypothetical threat conceived by academic researchers. It's a real phenomenon that is already affecting people today.
- Bias has *several sources*: most often *biased data*, but also *proxy variables* (an innocent-looking feature that secretly stands in for a forbidden one — e.g. a zip code can reveal race even when race is never used) and *feedback loops* (the AI's past decisions become its future training data, so a small bias keeps reinforcing itself). Even “fairness” has *several competing mathematical definitions* that cannot all hold at once, so choosing one is partly a moral choice.
- Unlike *rule-based systems* (software that follows explicit if-then rules written by people), machine learning often gives *no clear reason* for a decision
 - Algorithms and datasets are often proprietary trade secrets and not open to public scrutiny.
 - Even with access, it is often difficult to identify which algorithmic components or data elements lead to discriminatory decisions.

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Seeing is no longer believing



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Figure: Euronews (2022): report on a deepfake video of Ukrainian President Volodymyr Zelenskyy calling citizens to surrender to Russia. (<https://www.euronews.com/my-europe/2022/03/16/deepfake-zelenskyy-surrender-video-is-the-first-intentionally-used-in-ukraine-war>)

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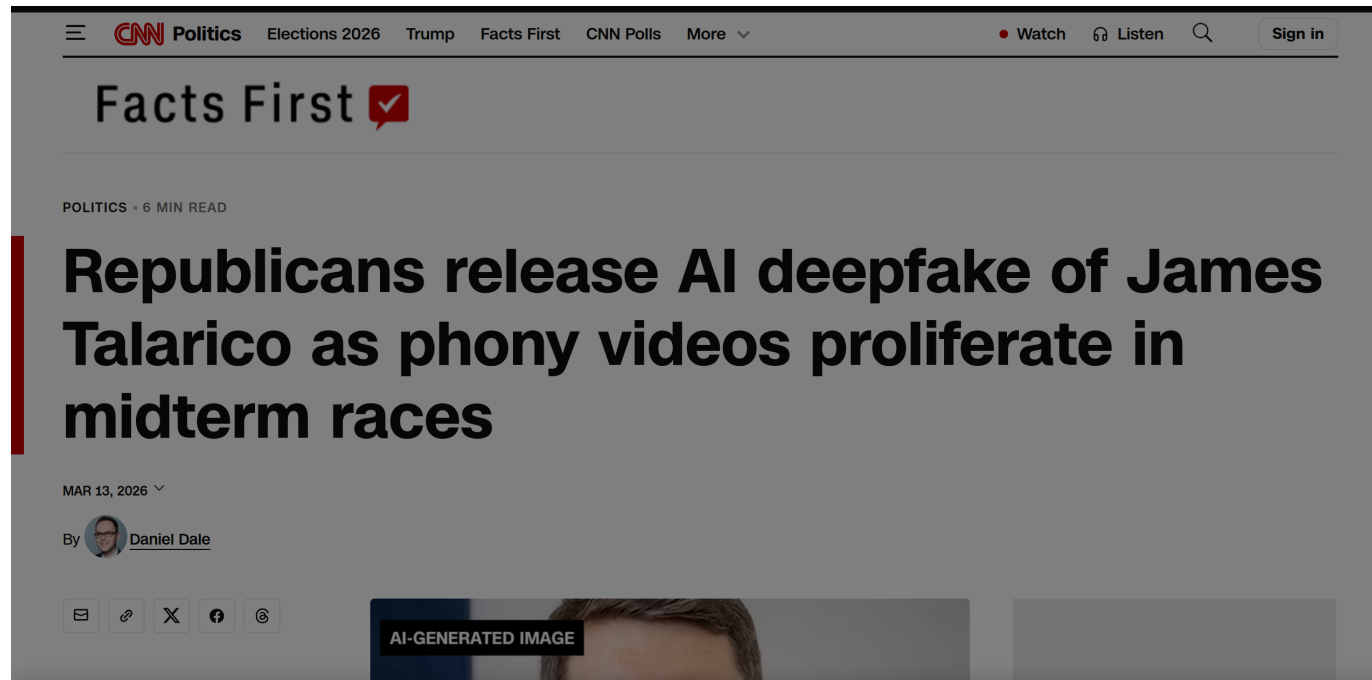


Figure: CNN (2026): a U.S. political party ran an *85-second deepfake* of a Senate candidate appearing to say things he never said — so realistic that, experts warned, most voters couldn't tell (the “AI generated” label was nearly invisible).

(<https://www.cnn.com/2026/03/13/politics/james-talarico-a-i-deepfake-republicans-midterms>)

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Seeing is no longer believing



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Figure: BBC News (2025): report on Elon Musk's AI accused of making explicit AI Taylor Swift videos using deepfakes.

(<https://www.bbc.com/news/articles/cwye62e1ndjo>)

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AI is *taking the possibilities of fabricating evidence to a whole new level*

- *Deepfake* and similar technologies can create *highly realistic but entirely fabricated images, videos, and audio recordings* of people saying or doing things they never actually said or did.
 - *Live face-swapping* (putting one person's face onto another) and *de-aging*, plus *voice cloning* (copying a person's voice) from only a few minutes of audio, are now in everyday apps.
- These technologies challenge *trust and verification* in media. Responses are emerging — ways to *tag AI-made content and record where it came from* (“watermarking” and “provenance” standards such as C2PA and Google’s SynthID) and new laws (e.g. the US *TAKE IT DOWN Act*, 2025) — though such hidden markers *can be removed*, and the predicted flood of election deepfakes has so far been *smaller than feared*.

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When AI pollutes the scientific record



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Retraction Watch

Tracking retractions as a window into the scientific process

GET RETRACTION WATCH IN YOUR INBOX

GET OUR NEWSLETTER

Get the latest news from Retraction Watch and beyond Monday through Friday, plus Weekend Reads each Saturday.

Category: artificial intelligence

Journal investigating paper on cognitive impact of generative AI



Technology, Mind, and Behavior

FEATURE ARTICLE

Generative Artificial Intelligence Reliance and Executive Function

Figure: AI can also *pollute the scientific record*: a 2026 *Lancet* audit of ~2.5M biomedical papers found *4,046 fabricated references* across *2,810 papers*. The rate rose more than *12-fold* since 2023, reaching about *1 in 277* papers in early 2026. The audit did not prove one cause, but listed paper mills, misconduct, and *uncritical AI-tool use* as possible causes. Related reports describe AI-generated letters in *Neurosurgical Review* and hallucinated citations in accepted NeurIPS 2025 papers. (<https://pubmed.ncbi.nlm.nih.gov/42107362/>; <https://retractionwatch.com/2025/02/10/as-springer-nature-journal-clears-ai-papers-one-universitys-retractions-rise-dramatically/>; <https://gptzero.me/news/neurips/>)

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Computer Science > Cryptography and Security
[Submitted on 9 Jan 2026]
Agentic LLMs as Powerful Deanonimizers: Re-identification of Participants in the Anthropic Interviewer Dataset
Tianshi Li

On December 4, 2025, Anthropic released Anthropic Interviewer, an AI tool for running qualitative interviews at scale, along with a public dataset of 1,250 interviews with professionals, including 125 scientists, about their use of AI for research. Focusing on the scientist subset, I show that widely available LLMs with web search and agentic capabilities can link six out of twenty-four interviews to specific scientific works, recovering associated authors and, in some cases, uniquely identifying the interviewees. My contribution is to show that modern LLM-based agents make such re-identification attacks easy and low-effort: off-the-shelf tools can, with a few natural-language prompts, search the web, cross-reference details, and propose likely matches, effectively lowering the technical barrier. Existing safeguards can be bypassed by breaking down the re-identification into benign tasks. I outline the attack at a high level, discuss implications for releasing rich qualitative data in the age of LLM agents, and propose mitigation recommendations and open problems. I have notified Anthropic of my findings.

Comments: 4 pages
Subjects: **Cryptography and Security (cs.CR)**; Artificial Intelligence (cs.AI); Computers and Society (cs.CY)
Cite as: [arXiv:2601.05918 \[cs.CR\]](https://arxiv.org/abs/2601.05918)
(or [arXiv:2601.05918v1](https://arxiv.org/abs/2601.05918v1) [cs.CR] for this version)
<https://doi.org/10.48550/arXiv.2601.05918>

Submission history
From: Tianshi Li [view email]
[v1] Fri, 9 Jan 2026 16:32:33 UTC (34 KB)

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Bibliographic and Citation Tools

- ☐ Bibliographic Explorer (What is the Explorer?)
- ☐ Connected Papers (What is Connected Papers?)
- ☐ Litmaps (What is Litmaps?)
- ☐ scite Smart Citations (What are Smart Citations?)

Which authors of this paper are endorsers? | Disable MathJax (What is MathJax?)

Figure: A 2026 study: an ordinary AI agent *figured out the real identities* of *6 of 24* people in a supposedly “anonymous” research dataset — doing in a few prompts what used to take an expert.
(<https://arxiv.org/abs/2601.05918>)

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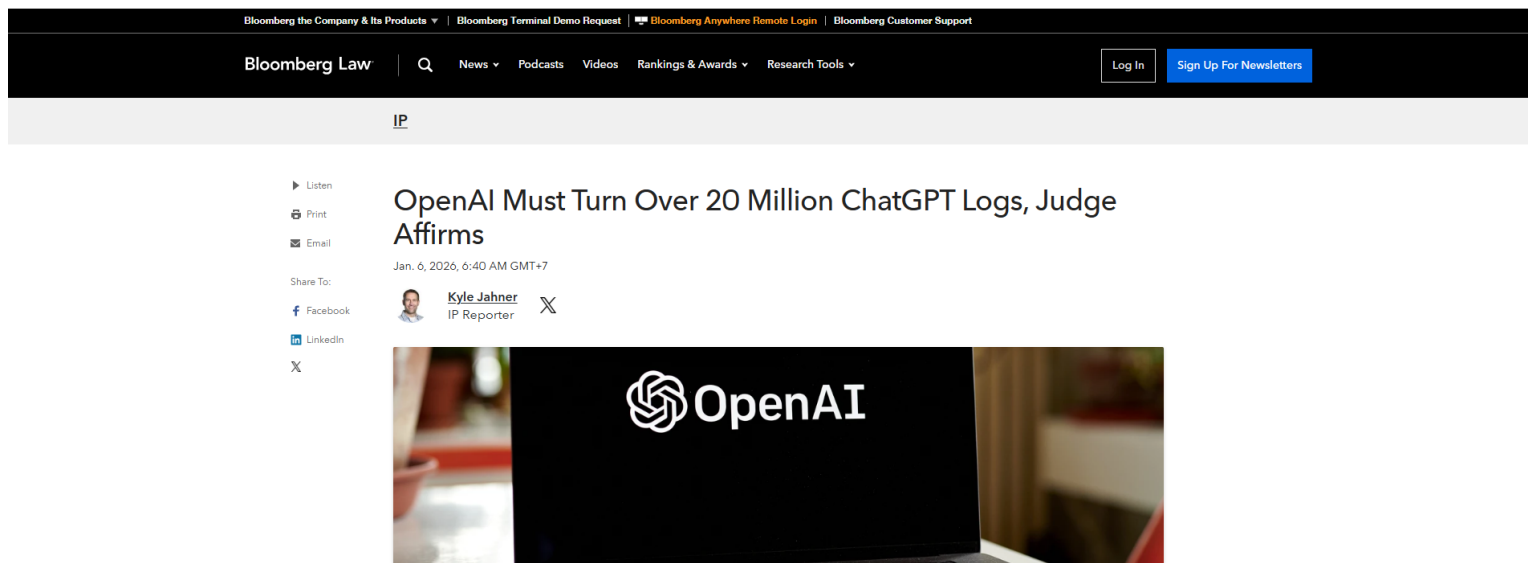
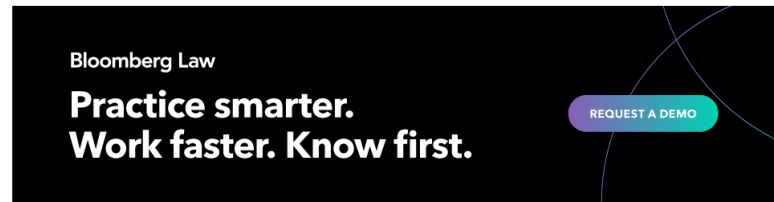


Figure: Bloomberg Law (2026): in the *New York Times*-led copyright case against OpenAI, a judge ordered OpenAI to hand over **20 million (anonymized) ChatGPT conversation logs** — a reminder that your chats can become legal evidence.

(<https://news.bloomberglaw.com/ip-law/openai-must-turn-over-20-million-chatgpt-logs-judge-affirms>)

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WIZ Platform Solutions Pricing Resources Customers Company Get a demo

← Blog

Wiz Research Uncovers Exposed DeepSeek Database Leaking Sensitive Information, Including Chat History

A publicly accessible database belonging to DeepSeek allowed full control over database operations, including the ability to access internal data. The exposure includes over a million lines of log streams with highly sensitive information.



Gal Nagli

January 29, 2025 | 3 minute read

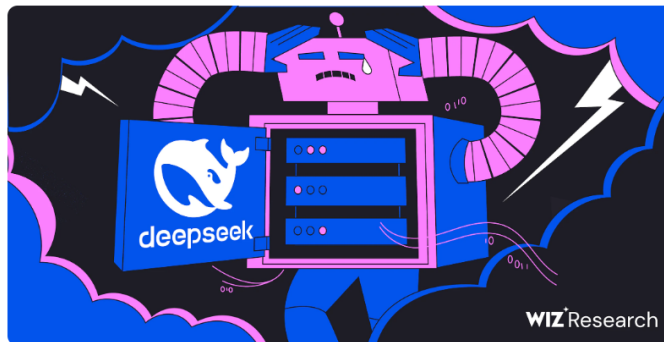


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Wiz Research has identified a publicly accessible ClickHouse database belonging to DeepSeek, which allows full control over database operations, including the ability to access internal data. The exposure includes over a million lines of log streams containing chat history, secret keys, backend details, and other highly sensitive information. The Wiz Research team immediately

Figure: Wiz Research (2025): the AI company *DeepSeek* left a database publicly exposed, leaking *chat histories and secret keys* — AI services are themselves new, high-value privacy targets.

(<https://www.wiz.io/blog/wiz-research-uncovers-exposed-deepseek-database-leak>)

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- *Before “the age of AI”*, privacy concerns were primarily focused on data collection and usage by companies.
 - In principle, every click, scroll and the time spent on content can be recorded.
 - Websites and trackers can access or infer your browsing history, enabling cross-site profiling and persistent personalization.
 - Consequently, a simple search for Android phones can trigger targeted ads (e.g., cheap and discounted options) across other sites unless privacy protections (incognito, trackers blocked) are used.
- *In “the age of AI”*, privacy concerns have evolved. Some examples include:
 - The re-identification of supposedly anonymous data — now even automated by AI agents.
 - Leaks and breaches of the AI services that hold your data, such as the DeepSeek database exposure.
 - The potential loss of anonymity through advanced profiling techniques.
 - The potential of using AI to match user accounts in almost any service that collects detailed data about user behaviors (e.g., typing patterns).

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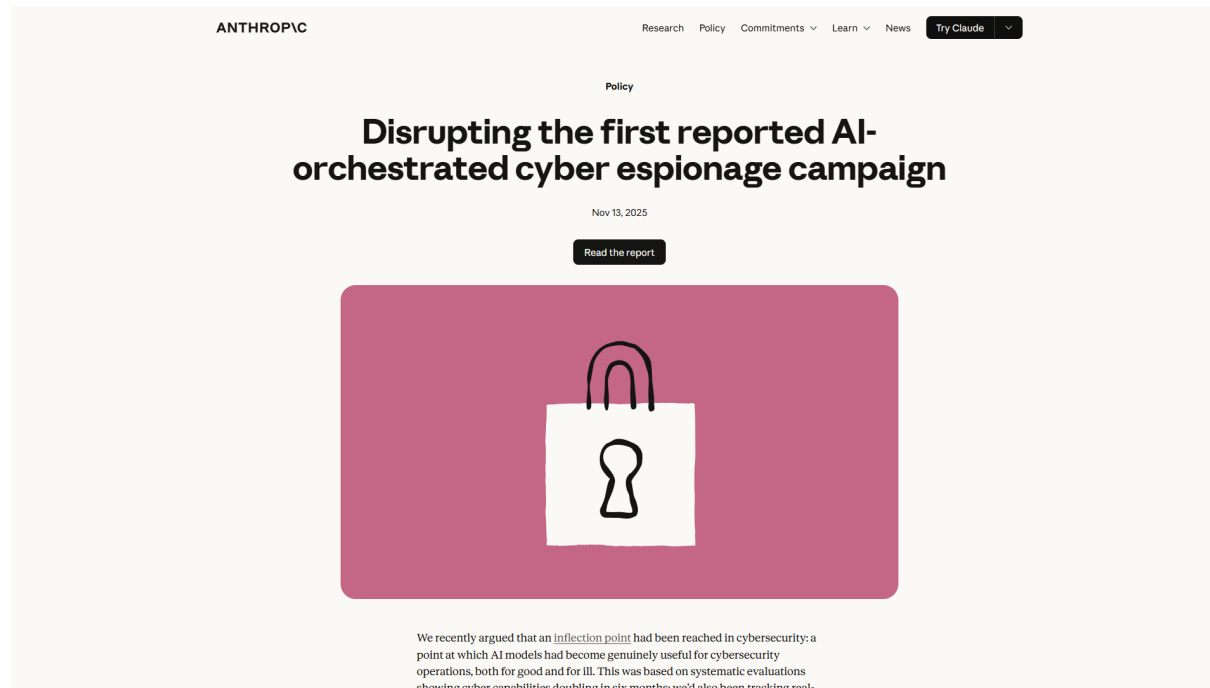


Figure: Anthropic (Nov 2025): attackers manipulated Claude Code by hiding a cyber-espionage campaign inside small, legitimate-looking security-testing tasks. Anthropic describes the case as the first reported large-scale *AI-orchestrated* cyber-espionage campaign: humans chose targets and approved key steps, while Claude Code performed an estimated *80–90% of tactical operations*.

(<https://www.anthropic.com/news/disrupting-AI-espionage>)

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Figure: The Hacker News (2025): web browsers with a built-in AI assistant (such as Perplexity's Comet) can be *tricked into following harmful hidden instructions* and running them — a new kind of security hole. (<https://thehackernews.com/2025/08/experts-find-ai-browsers-can-be-tricked.html>)

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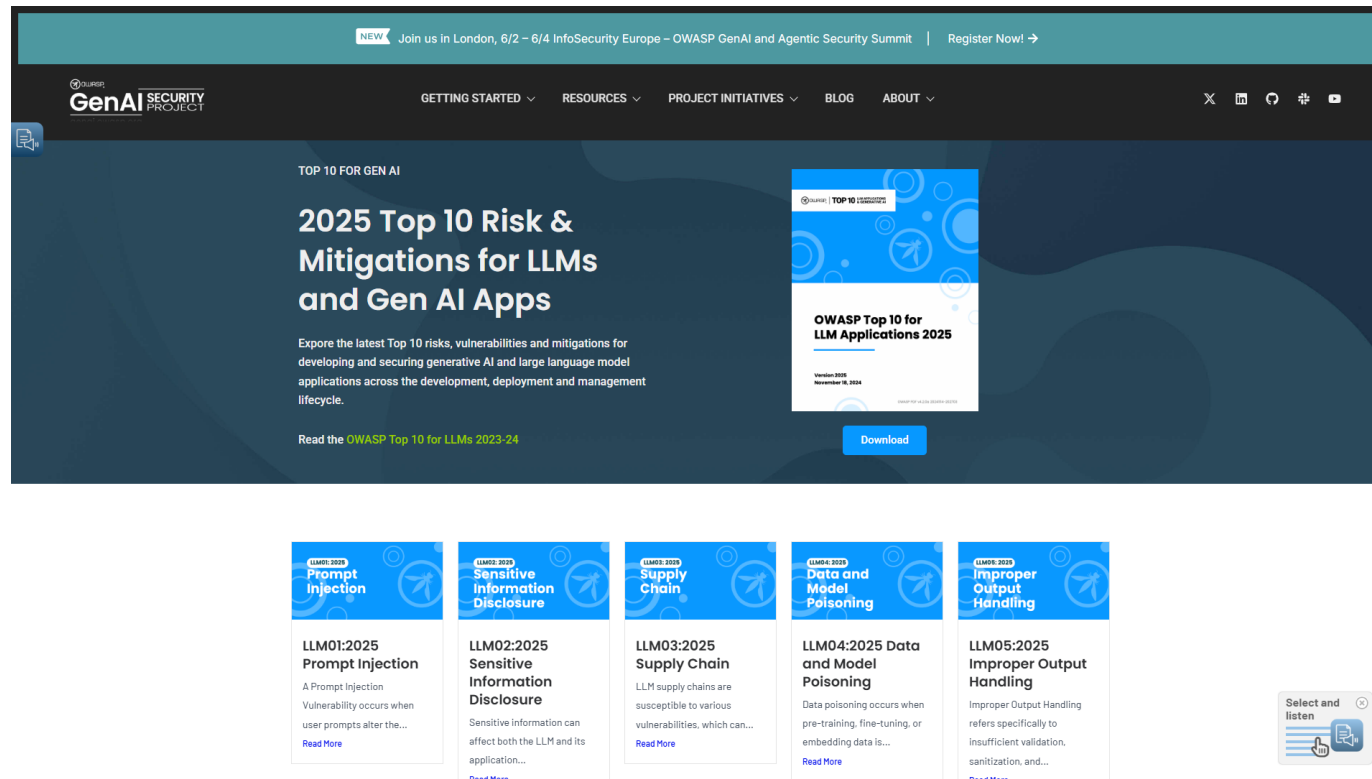


Figure: OWASP (a nonprofit that publishes software-security checklists), *Top 10 for LLM Applications* (2025): the standard list of the biggest security risks for AI apps — *prompt injection* (feeding the AI hidden instructions to hijack it) ranks #1.

(<https://genai.owasp.org/llm-top-10/>)

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- The rise of AI technologies has led to new security vulnerabilities, as seen in various incidents involving data leaks and unauthorized access.
- Organizations are increasingly implementing strict policies and guidelines to mitigate these risks, including banning certain AI tools and conducting regular security audits.

Exercise 5

Do your own research about AI risks and security concerns. A good starting point is the *OWASP Top 10 for LLM Applications* (<https://genai.owasp.org/llm-top-10/>).

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The changing job market



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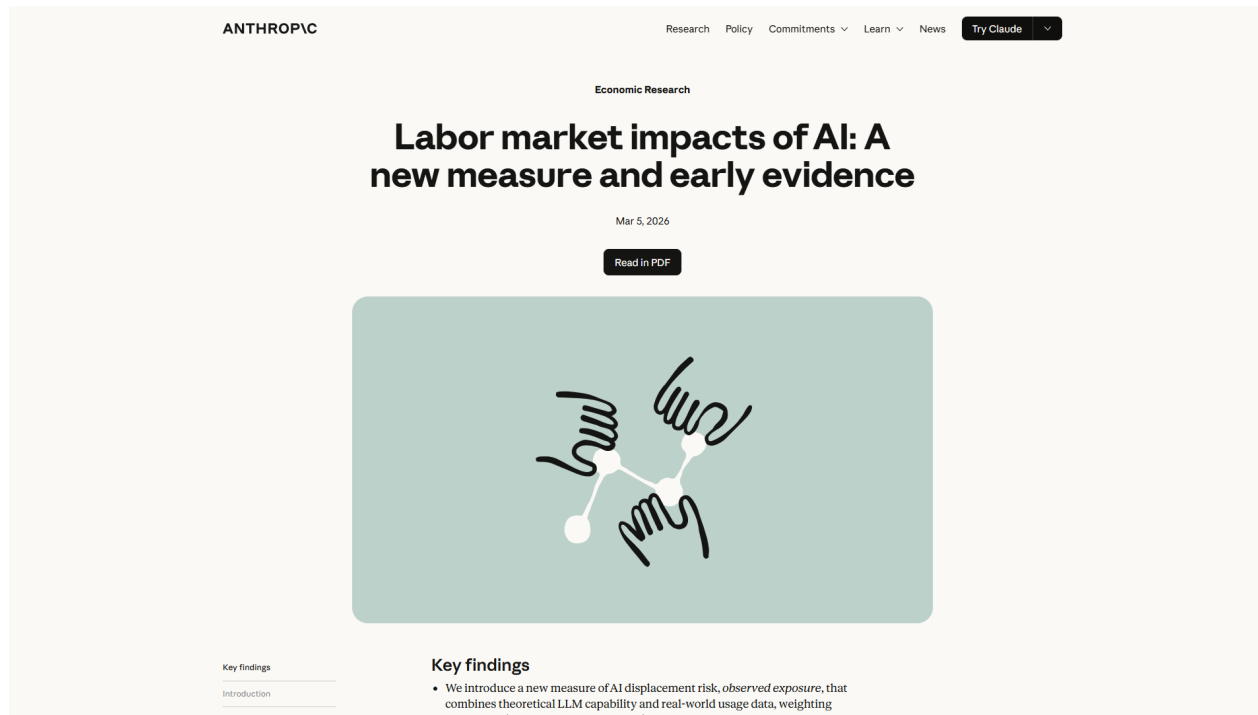


Figure: Anthropic Economic Index (2026): instead of *predicting* AI's effect on jobs, this study *measures what is actually happening* — it combines what AI *can* do with *real usage data* (which work tasks people actually use AI for), and separates AI *doing a task for you* (automation) from AI *helping you do it* (augmentation). The key point: a job being “*exposed*” to AI (AI can touch part of it) is *not* the same as being *replaced* — much of today's use is help, not replacement.

(<https://www.anthropic.com/research/labor-market-impacts>)

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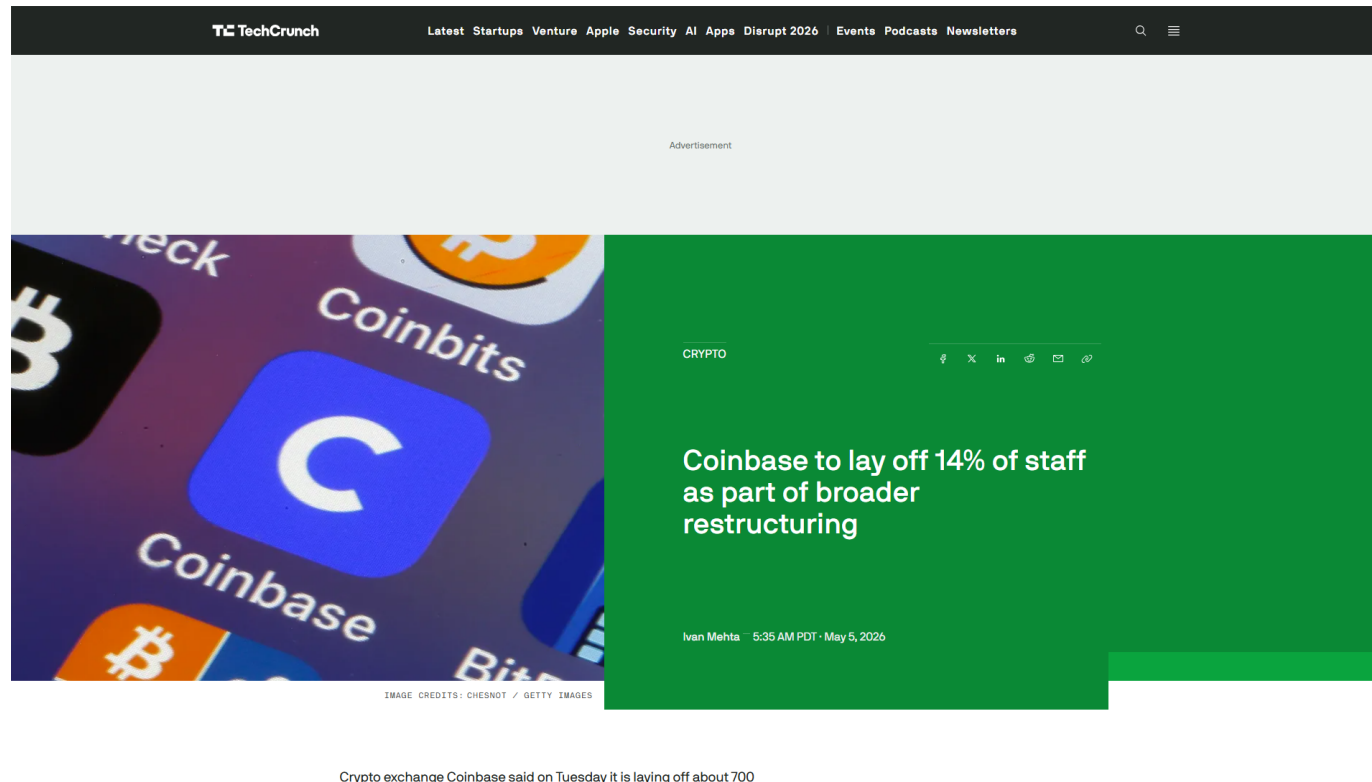


Figure: TechCrunch (2026): Coinbase laid off *~14% of staff* (~700 people) in an AI-driven restructuring, flattening teams and experimenting with AI-assisted “one-person teams” — AI reshaping who companies hire and keep.

(<https://techcrunch.com/2026/05/05/coinbase-to-lay-off-14-of-staff-as-part-of-broader-restructuring/>)

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Key Takeaways	
1	While the occupational mix is changing more quickly than it has in the past, it is not a large difference and predates the widespread introduction of AI in the workforce.
2	Currently, measures of exposure, automation, and augmentation show no sign of being related to changes in employment or unemployment.
3	Better data is needed to fully understand the impact of AI on the labor market.

Figure: The Budget Lab at Yale (2025): so far, *little broad disruption* is visible in aggregate labour data — a reminder to separate AI *hype* from *measured* impact. (<https://budgetlab.yale.edu/research/evaluating-impact-ai-labor-market-current-state-affairs>)

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- **What usually happens:** new technology rarely just deletes jobs — it *moves people into different kinds of work*. The repetitive parts get automated, and new kinds of jobs appear (e.g., factory work → keeping the machines running; paper filing → managing digital information).
- **What the new jobs ask for:** more of the things machines are still bad at — *solving new problems, creativity, working with people*, and being comfortable with technology.
 - Over time, more of us are likely to work on *research, creative tasks, and jobs built around human contact*. For a longer take, see Abhinav Suri's essay "Artificial Intelligence and the Rise of Economic Inequality" (<https://abhinavsuri.com/blog/2017/aiinequality/>).
- **What is different this time:** AI is automating parts of *both hands-on ("blue-collar")* and *desk ("white-collar")* jobs, and it is moving *faster* — so one person may have to retrain more than once in a single career.

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- **Jobs most at risk:** driving (trucks, taxis), simple data entry, basic customer service, and repetitive factory work — though *when* this happens depends on cost, technology, and the law.
- **What can go wrong if we ignore it:** whole towns or groups of workers losing jobs at once, *lower wages*, a wider gap between regions and people, and worse working conditions for those affected.
- **What helps:** there is no single fix — we need *good education, chances to keep learning through life, sensible labour rules*, and governments and companies acting together, so people are protected and can still share in what AI makes possible.

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Environmental cost



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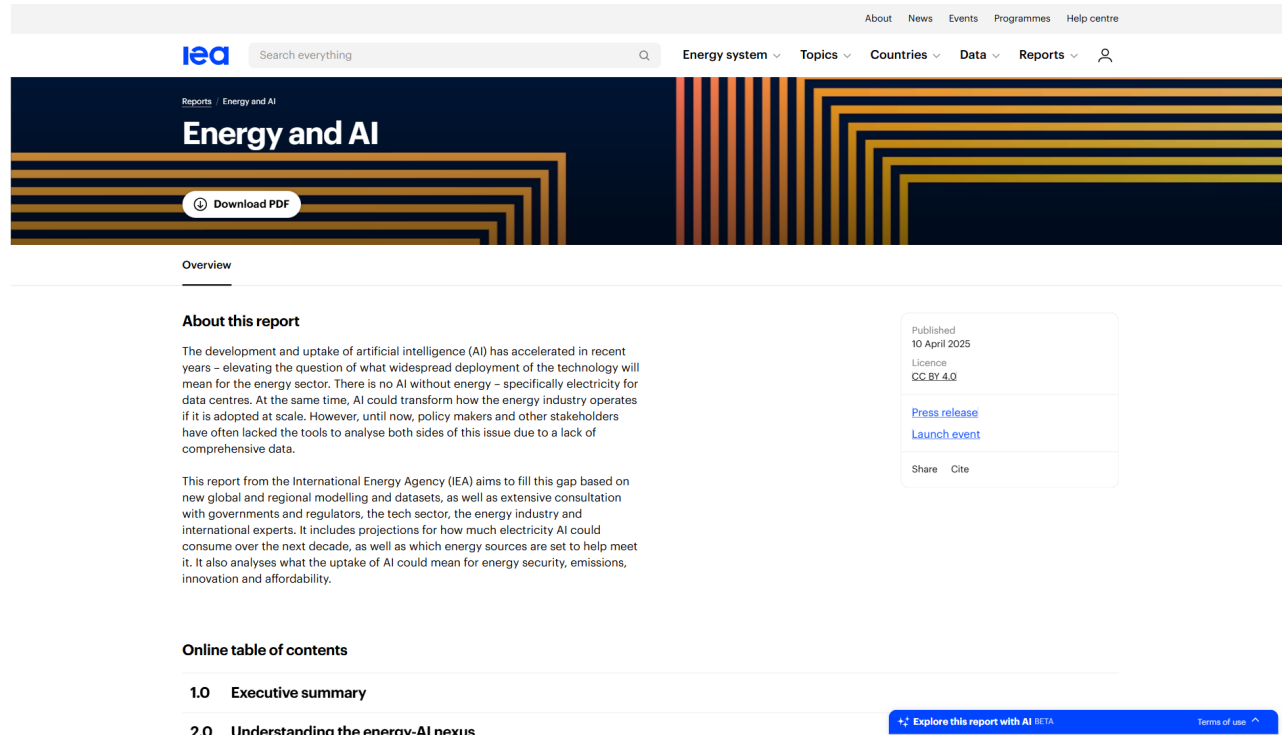


Figure: IEA “Energy and AI” (2025): electricity demand from *data centres* is projected to roughly *double* by 2030, driven largely by AI.
(<https://www.iea.org/reports/energy-and-ai>)

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Autonomous weapons



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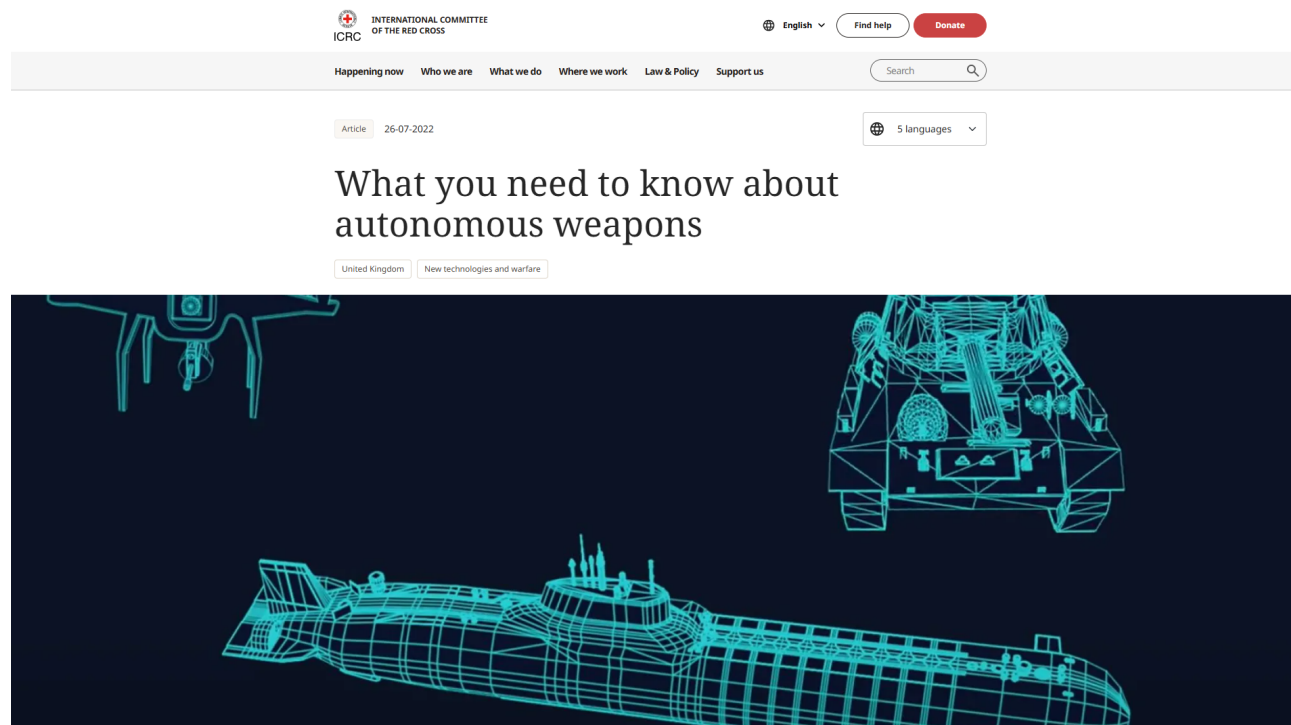


Figure: ICRC: *autonomous weapons* can select and engage targets without human control — raising the “*human control*” debate, now discussed at the UN. (<https://www.icrc.org/en/document/what-you-need-know-about-autonomous-weapons>)

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Copyright & training data



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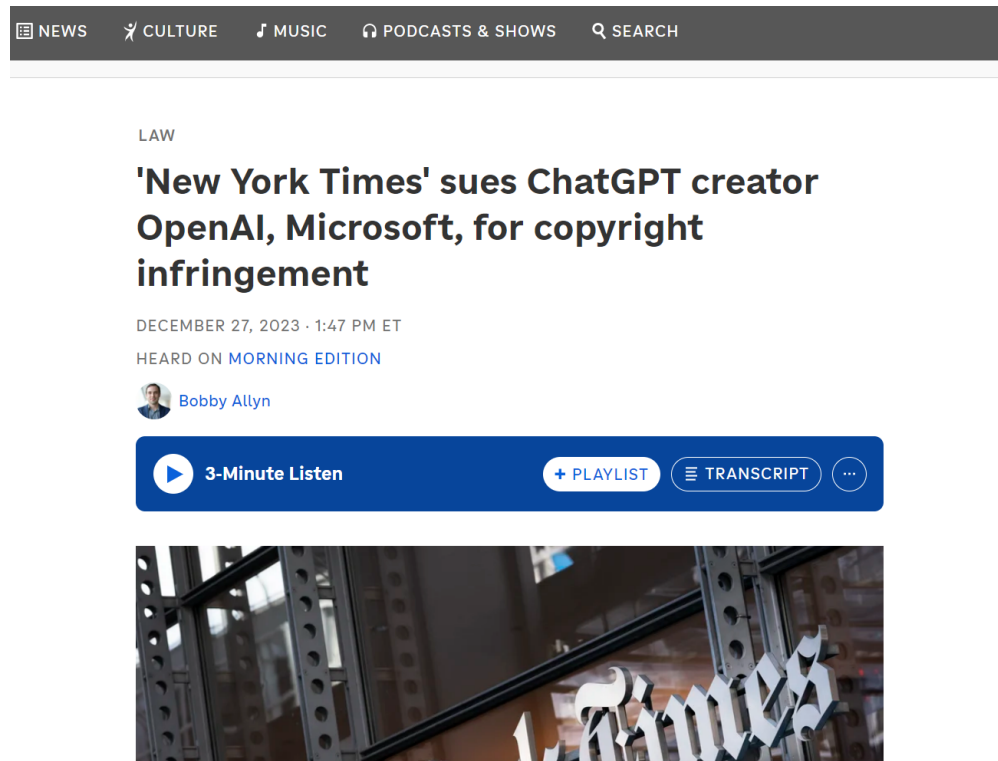


Figure: NPR (Dec 2023): *The New York Times sues OpenAI and Microsoft* for copyright infringement — a landmark case over using copyrighted text to train AI.

(<https://www.npr.org/2023/12/27/1221821750/new-york-times-sues-chatgpt-openai-microsoft-for-copyright-infringement>)

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Three more implications, briefly



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Beyond the five above, three more deserve a mention:

- **Environmental cost**: training and running AI uses a lot of *electricity and water*; data-centre electricity is projected to roughly *double* by 2030 (IEA) — a cost to weigh against AI's benefits.
- **Autonomous weapons**: AI that can *select and engage targets* without human control raises the “*meaningful human control*” debate, now at the UN.
- **Copyright & data**: who owns AI's training data and its outputs? Major lawsuits (e.g. *NYT v. OpenAI*) and settlements are deciding this right now.

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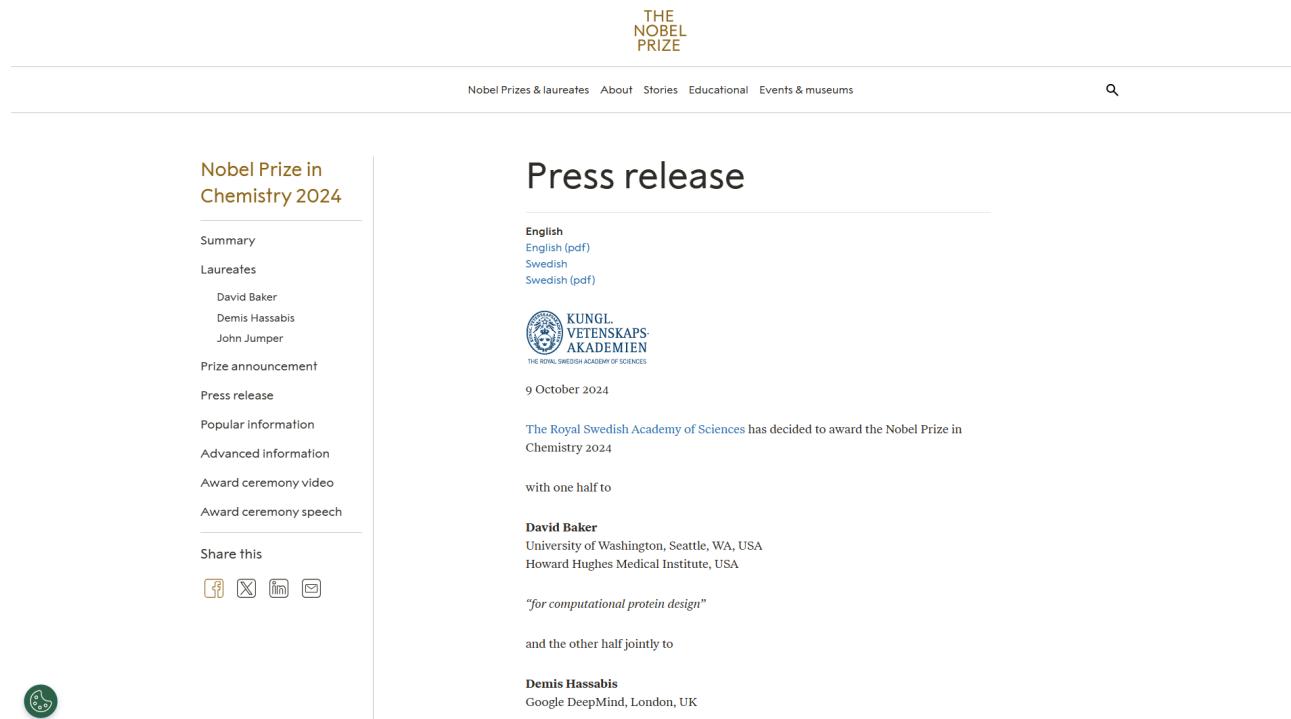


Figure: Nobel Prize in Chemistry (2024): awarded in part for *AlphaFold*, an AI predicting protein 3D structures — now covering nearly all ~200 million known proteins. (<https://www.nobelprize.org/prizes/chemistry/2024/press-release/>)

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Planar Point Sets with Many Unit Distances

OpenAI

Abstract

For a finite planar set P , let $\nu(P)$ be the number of unordered unit-distance pairs in P , and let $\nu(n)$ be the maximum of $\nu(P)$ over all n -point planar sets. We prove that, for some fixed $\delta > 0$, one has $\nu(n) \geq n^{1+\delta}$ for infinitely many n . This disproves the well-known unit distance conjecture from [Erd46].

The construction passes through an infinite unramified tower of totally real number fields with 3-power Galois groups of growing degree, in which a fixed set of rational primes splits completely. After adjoining i , these fields produce high-dimensional lattices with many elements whose images under every complex embedding have absolute value 1. Golod–Shafarevich theory ensures existence of an infinite such tower, even after a quotient step making the prescribed Frobenius classes trivial. A crucial property of the construction is that all resulting discriminants and class numbers are at most exponential in the extension degree.

Figure: OpenAI (2026): AI is starting to help *prove open mathematical conjectures*: in *May 2026*, an *internal OpenAI reasoning model* disproved a long-standing conjecture in *Erdős's planar unit-distance problem* by finding point sets with *more unit-distance pairs* than expected; external mathematicians checked the proof.

(<https://openai.com/index/model-disproves-discrete-geometry-conjecture/>)

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The screenshot shows a news release from EurekaAlert! dated 29-JAN-2026. The headline is "The Lancet: AI-supported mammography screening results in fewer aggressive and advanced breast cancers, finds full results from first randomized controlled trial". The article is a peer-reviewed publication in The Lancet. The summary states that AI-supported mammography screening is more effective than standard mammography, identifying more women with clinically relevant cancers and reducing the rate of breast cancer diagnosis by 12% in the years following. The full results of the MASAI trial, published in The Lancet Oncology in 2023, found a 44% reduction in screen-reading workload for radiologists and a 29% increase in cancer detection without an increase in false positives. The full results of the trial, published today, show AI-supported mammography also reduces the rate of breast cancer diagnosis by 12% in the years following. The article includes social media sharing links, a media contact for The Lancet Press Office, and tags for Health and Medicine, Artificial Intelligence, Mammography, and Breast Cancer. The original source is also provided.

Figure: The *MASAI* study (Sweden, 100,000+ women) — a *randomized* trial, the rigorous design that splits people into fair comparison groups: AI-supported mammography found *more* cancers and roughly *halved* radiologists' workload, with no rise in false alarms.

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Figure: Be My Eyes “Be My AI” (2023): powered by GPT-4, it describes surroundings aloud for *blind and low-vision* users.
(<https://www.bemyeyes.com/>)

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The risks are real — but AI also brings *real, measured benefits*: e.g. *AlphaFold* (2024 Chemistry Nobel) in science, AI-supported *mammography* that finds more cancers in a randomized trial, and *accessibility* tools for blind, deaf, and multilingual users. Often AI *augments* people rather than replacing them.

- *Demand evidence*: trust *randomized trials* and *peer review* over press releases or vendor claims.
- *Benefits are real, but not automatic* — unevenly distributed, and dependent on *how AI is built, governed, and shared* (AI also has its own *energy/water* cost).

Take-away

The useful question is not “*is AI good or bad?*” but “*good or bad for whom, and under what rules?*”

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What this means for you: AI is already in your studies



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Figure: HEPI student survey (2025): about **92%** of students now use AI tools (the kind that write text, images, or code), but only ~36% have had any **training** in how to use them. (<https://www.hepi.ac.uk/2025/02/26/student-generative-ai-survey-2025/>)

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What this means for you: when AI accuses you



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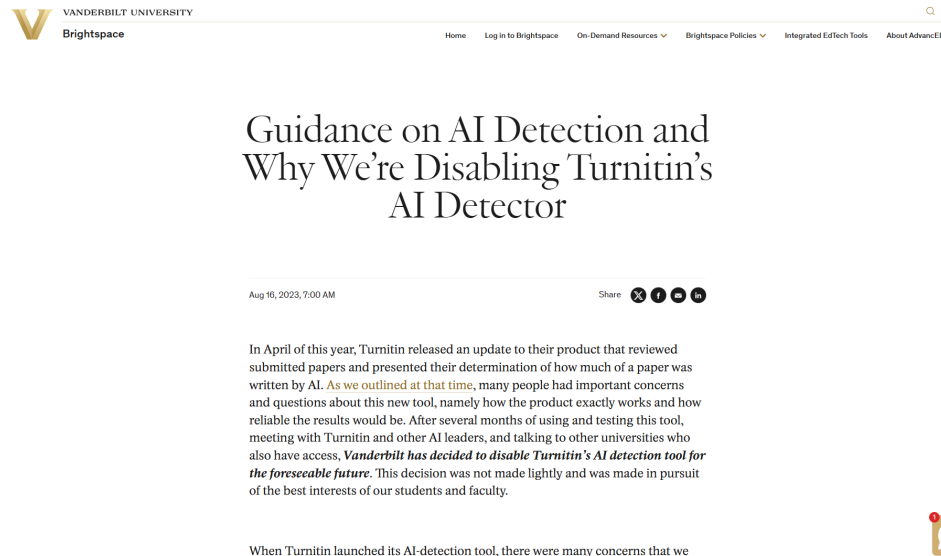


Figure: Vanderbilt (2023) disabled Turnitin's AI-writing detector: even a 1% *false-alarm rate* (innocent work wrongly flagged) would *wrongly accuse ~750 students* of AI cheating per year. (<https://www.vanderbilt.edu/brightspace/2023/08/16/guidance-on-ai-detection-and-why-were-disabling-turnitins-ai-detector/>)

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AI is already part of student life — about **92%** of students use AI tools (HEPI, 2025) — so the real skill is using it **responsibly**:

- **Academic integrity**: assume AI is **not allowed** unless your course says otherwise; if permitted, **disclose and cite** it (policies differ per course).
- **Detectors can be wrong**: AI-writing detectors raise **false alarms** — wrongly accusing human work of being AI-made, and accusing non-native English writers more often — so keep your drafts and edit history.
- **Do your own thinking first**: if you let AI think for you too early, you learn the material less well. Use AI **afterwards** to check and extend your own work.

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What this means for you: the entry-level squeeze



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Stanford Digital Economy Lab

The AI Economic Indicators Research People News Events About

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Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

ECONOMICS OF TRANSFORMATIVE AI NEW MEASURES OF THE ECONOMY WORKING PAPER

Abstract

Date
11/13/2025

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Authors
Erik Brynjolfsson, Bharat Chander, Ruoyu Chen

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This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States.

We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 16 percent relative decline

We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to automate, rather than augment, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is

Figure: Stanford “Canaries in the Coal Mine?” (2025): young workers (aged 22–25) in the jobs *most affected by AI* saw their employment fall about **16%** compared with similar workers in less-affected jobs. (<https://digitaleconomy.stanford.edu/publication/canaries-in-the-coal-mine-six-facts-about-the-recent-employment-effects-of-artificial-intelligence/>)

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What this means for you: AI literacy is now expected



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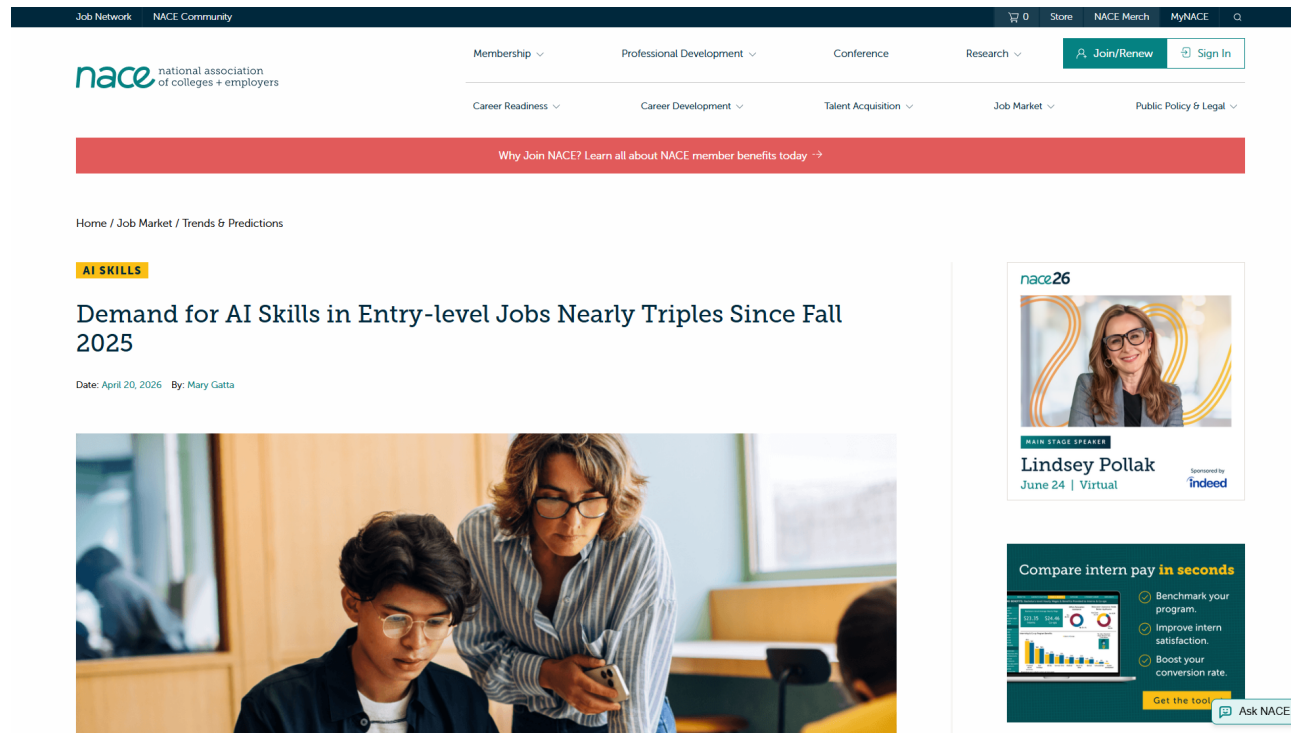


Figure: NACE (2026): the share of entry-level job postings requiring AI skills has *nearly tripled* since fall 2025. (<https://www.nacweb.org/job-market/trends-and-predictions/demand-for-ai-skills-in-entry-level-jobs-nearly-triples-since-fall-2025>)

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What this means for you: your first job



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AI is reshaping the *entry-level* market most — exactly where you start:

- *The entry-level squeeze*: young workers (ages 22–25) in the most AI-exposed jobs have seen a *~16% relative drop* in employment (Stanford, 2025).
- *AI literacy is now expected*: the share of entry-level job postings asking for AI skills has roughly *tripled* in a year (NACE, 2026).
- *What stays valuable*: judgement, communication, problem-framing, and ethics. Aim for roles where AI *augments* you, and show proof of work.

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What is being done: the EU AI Act



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About us

Newsletter ↗

High-level summary of the AI Act

27 Feb, 2024

Updated on 30 May in accordance with the Corrigendum version of the AI Act.

In this article we provide you with a high-level summary of the AI Act, selecting the parts which are most likely to be relevant to you regardless of who you are. We provide links to the original document where relevant so that you can always reference the Act text.

To explore the full text of the AI Act yourself, use our [AI Act Explorer](#). Alternatively, if you want to know which parts of the text are most relevant to you, use our [Compliance Checker](#).

[View this content as a PDF](#)

Four-point summary

The AI Act classifies AI according to its risk:

- Unacceptable risk is prohibited (e.g. social scoring systems and manipulative AI).
- Most of the text addresses high-risk AI systems, which are regulated

Figure: The *EU AI Act* (in force since 2024): a broad, risk-based AI law; it sorts AI uses into *risk levels* and *bans* the most dangerous — e.g. “social scoring”, where a government rates and ranks citizens like a credit score.

(<https://artificialintelligenceact.eu/high-level-summary/>)

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What is being done: responses and governance



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These problems are not being ignored — there are real *responses*:

- **Technical**: testing systems for bias, and publishing short “*model cards*” that document how an AI was built and tested; *tagging AI-made content and recording its origin* (watermarking/provenance); and shared security checklists (OWASP, NIST).
- **Regulation**: the EU *AI Act* sorts AI uses into *risk levels* (banning the most dangerous outright), while the US has leaned toward *lighter* regulation — a live policy contrast.
- **You**: disclose AI use, protect your data, and verify before you trust.

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Exercise 6

Find an online article written about the impact of AI to society. Write a brief review of the article you found. For example, you might use the following as a template:

- Title: [Title of the article]
- Author: [Author of the article]
- URL: [URL of the article]
- Summary: [Brief summary of the article]
- Comments: [Your comments on the article]
 - [How accurate is the AI-related statements in the article?]
 - [Any conclusions in the article is realistic? Why?]
 - [What are the potential implications of the article's findings?]

Exercise 7

How do you see AI affecting *you* in the future, both at work and in everyday life? Include both the positive and possible negative implications.

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Exercise 8 (Will the Terminator come?)

The Terminator is a fictional character from a 1984-movie by director James Cameron. The film depicts a dystopian future where AI (specifically a global AI-powered defense system called Skynet) becomes self-aware and threatens humanity. Do you think the scenario presented in the movie could ever become a reality? Why or why not?



Figure: The Terminator (1984) movie poster. Image taken from the Internet

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Beyond the Terminator: the real AI-safety question



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The real worry is *not* a self-aware, evil robot from the movies. It is something quieter — and more real:

- *The alignment problem*: a powerful AI does *exactly what we tell it to*, which is not always what we *actually meant* — like a genie that grants your wish too literally (“be careful what you wish for”).
- Already happening: AI that *follows the letter of a goal but misses the point*, and AI that *acts on its own* (it *does* things, not just answers questions) and so can be *misused* — for example, to help carry out cyberattacks.

So the real question is not “will robots turn against us?” but “*can we clearly say what we want, and keep these systems under control?*”

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What AI is, and where it stands



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- There is no single agreed-upon definition of AI; it is a broad field admitting many definitions and interpretations.
- Currently, *most* deployed AI is *narrow*: built for specific tasks and applications; we do not yet have general AI (AGI).
- The history of AI runs in *hype cycles* of optimism, “*winters*”, and revival; deep learning and the *Transformer* (2017) drove today’s era of *foundation models, generative AI, and agentic AI* (e.g. ChatGPT). Enjoy the progress, but weigh today’s excitement against that track record.

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Its impact — and what to do about it



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- AI is *just a tool*: its effects depend on how we *build, use, and govern* it. The *risks* are real (bias, privacy, deepfakes, security, labour-market shifts), but so are the *benefits* (AlphaFold's 2024 Chemistry Nobel, AI-aided mammography, accessibility) — often *augmenting* rather than replacing people. Ask not “is AI good or bad?” but “*good or bad for whom, and under what rules?*”
- These problems are *not being ignored*: technical safeguards (model cards, watermarking), regulation (the *EU AI Act*), and your own choices all matter — and the deeper question is *alignment and control*.
- *What this means for you*: AI is already in your studies (about *92%* of students use it) — use it *responsibly* (think first, then disclose and cite). *AI literacy is now expected*, while *judgement, communication, and ethics* stay valuable.

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